Gain and Bandwidth Enhancement of Compact GNSS L1 Antenna using Rectangular DGS and Symmetric Patch Slotting

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***Abstract*—This paper proposes a compact embedded microstrip patch antenna designed for the Global Navigation Satellite System (GNSS) L1 frequency band (1.575 GHz). The antenna is fabricated on a low-cost FR-4 substrate with a dielectric constant of 4.4 and a thickness of 1.6 mm. To address the gain limitations and surface wave losses typically associated with lossy substrates, the design incorporates a Defected Ground Structure (DGS) featuring rectangular slots, along with four symmetric corner slots on the radiating patch. These geometric modifications enhance radiation efficiency by suppressing surface waves and extending the effective electrical length to achieve miniaturization. The proposed antenna was simulated using Ansys HFSS, demonstrating a peak gain of 2.38 dB and an S_{11} of -20.82 dB. A physical prototype was fabricated and tested, yielding a measured S_{11} of -22.186 dB at a resonant frequency of 1.632GHz. The results validate the efficacy of the proposed design in delivering high-gain performance within a compact form factor, making it suitable for embedded navigation applications.**

Keywords: Microstrip Patch Antenna, GNSS L1, Defected Ground Structure (DGS), Miniaturization, FR-4, Ansys HFSS.

I. INTRODUCTION

The demand for integrated positioning systems in portable electronics has led to the development of compact, cost-effective antennas for the Global Navigation Satellite System (GNSS). While the L1 frequency band (1.575 GHz) is the standard for civilian use, designing high-performance antennas for this band on low-cost FR-4 substrates is challenging due to surface wave losses. However, FR-4 poses substantial difficulties at RF frequencies; its high dielectric constant ($\varepsilon_r \approx 4.4$) and loss tangent promote the generation of surface waves. These waves propagate along the substrate boundary rather than radiating, resulting in edge scattering and degraded gain. Consequently, significant research has been directed toward mitigating these substrate losses through geometric modifications. Extensive literature indicates that manipulating the ground plane and patch geometry can effectively suppress these losses.

In the specific domain of GPS applications, Ayad *et al.* [1] utilized notched slots to achieve a gain of 2.27 dB on a single-layer substrate. To further enhance radiation efficiency in such designs, Defected Ground Structures (DGS) have been widely adopted. Rusdiyanto *et al.* [2] demonstrated that DGS significantly improves bandwidth and gain by modifying the current distribution. The utility of this technique extends beyond simple gain enhancement; Al-Falahy *et al.* [3] utilized it to reduce mutual coupling in cylindrical arrays, demonstrating its robustness. Furthermore, Alam *et al.* [4] confirmed the effectiveness of ground modifications specifically on epoxy woven glass (FR-4) materials, validating the approach for lossy substrates. These engineered ground plane techniques, comprehensively reviewed by Dwivedy [5], function as spatial filters that suppress surface wave propagation.

Subsequent research has focused on further miniaturization. Veisee *et al.* [6] proposed novel compact DGS shapes to minimize lateral scattering, and Majid *et al.* [7] explored left-handed metamaterials to achieve comparable efficiency improvements. More recently, Wani *et al.* [8] analyzed methods for enhancing gain in ultra-wideband scenarios. However, achieving high gain often involves complex structures; for instance, Li *et al.* [9] employed a stacked patch configuration with high-permittivity ceramics to reach 2.93 dBi, and Savarkar *et al.* [10] developed geodetic grade antennas for the L5 band.

In the specific domain of GNSS antennas, recent works illustrate the trade-off between complexity and performance. This paper proposes a compact, high gain microstrip patch antenna optimized for the GNSS L1 band which addresses these challenges without resorting to complex multi-layer structures. Building on the fundamental dimensions explored by Ayad et al. [1] , our design introduces a combination of a rectangular DGS to enhance radiation efficiency and symmetric corner slots for miniaturization. By strategically disrupting the surface current distribution, the proposed antenna achieves resonance at 1.575 GHz with a peak gain of 2.38 dB on a single

layer of lossy FR-4. The design is validated through full-wave simulations in Ansys HFSS and experimental measurements, demonstrating that strategic geometric defects can effectively overcome the inherent limitations of low-cost substrates.

II. ANTENNA DESIGN

The proposed antenna is designed for the GNSS L1 (1.559-1.610 GHz) band which has a centre frequency of 1.575 GHz. The structure is implemented on a cost-effective FR-4 substrate with a relative permittivity (ε_r) of 4.4, a thickness (h) of 1.6 mm.

A. Radiating Element and Miniaturization

The radiating element is based on a rectangular microstrip patch configuration. The initial geometric dimensions were adopted from the design proposed in [1], utilizing a patch size of 58.5×44.2 mm^2 and a substrate size of 95×82 mm^2. To achieve the miniaturization required for embedded applications, four symmetric square slots (5×5 mm^2) were introduced at the corners of the radiating patch. These slots perturb the surface current distribution, forcing the current to traverse a longer path. This modification effectively increases the electrical length of the antenna, allowing it to resonate at the target GNSS L1 frequency (1.575 GHz) without increasing its physical footprint.

B. Feed Mechanism

An inset feed technique was employed to achieve precise impedance matching. A notch with length L_n and width W_n was cut into the patch, recessing the feed point to a location where the input impedance matches the 50Ω transmission line. This minimizes reflections and optimizes the return loss.

C. Defected Ground Structure (DGS)

To overcome the gain limitations inherent to lossy FR-4 substrates, a Defected Ground Structure (DGS) was implemented. Two rectangular slots were etched into the top corners of the ground plane. Based on suppression principles described in literature [2], these defects modify the boundary conditions of the ground plane. The structure acts as a band-stop filter that suppresses surface wave propagation and prevents edge scattering. This mechanism directs electromagnetic energy more efficiently into the main lobe, directly contributing to enhanced antenna gain.

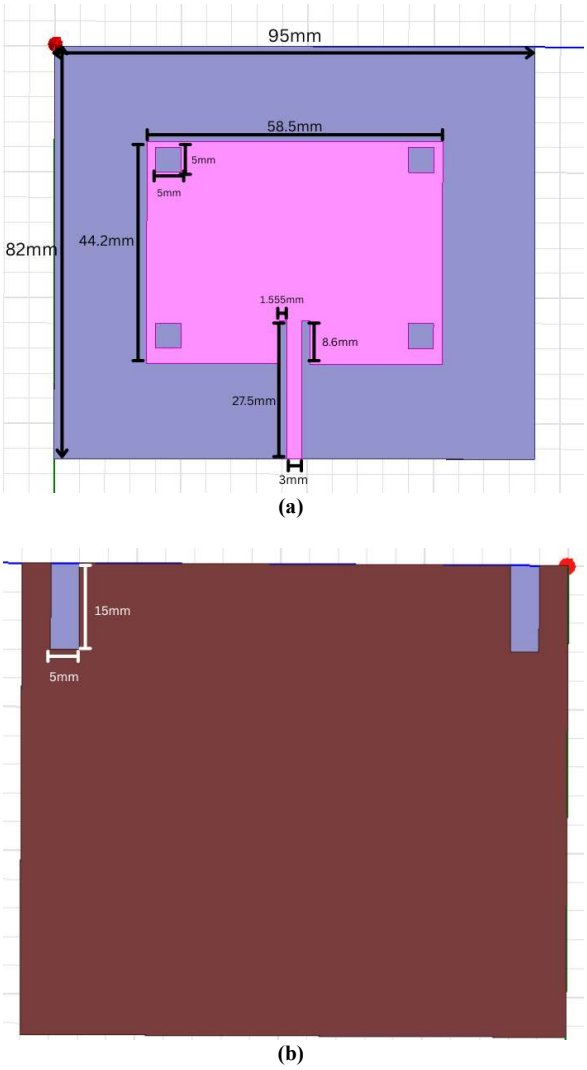


Fig. 1. Geometry of the proposed antenna: (a) Top view showing corner slots and inset feed; (b) Bottom view showing Defected Ground Structure (DGS).

Table 1: Antenna Dimensions

Parameter	Dimensions (mm)
Substrate Length (L_{sub})	82
Substrate Width (W_{sub})	95
Patch Length (L_p)	44.2
Patch Width (W_p)	58.5
Patch Slot Size	5×5
Feed Line Length (L_f)	27.5
Feed Line Width (W_f)	3
Notch Length (L_n)	8.6
Notch Width (W_n)	1.555
DGS Slot Size	15×5

III. RESULTS AND DISCUSSION

The performance of the proposed antenna was validated through a comparative analysis of simulation stages followed by experimental measurements of the fabricated prototype.

A. Design Evolution (Simulation)

To quantify the impact of the proposed geometric modifications, the design was simulated in two iterative stages. The antenna was designed and simulated using Ansys High Frequency Structure Simulator (HFSS), an industry-standard tool utilizing the Finite Element Method (FEM) for 3D electromagnetic analysis.

The simulation setup employed a radiation boundary box to simulate a vacuum environment.

To accurately analyse the resonance and bandwidth characteristics, a Fast Frequency Sweep was configured with the following parameters:

- **Start Frequency:** 1.0 GHz
- **Stop Frequency:** 2.0 GHz
- **Step Count:** 301 points
- **Loss Tangent (tan δ):** 0.02

This high-resolution sweep ensured the precise identification of the resonant frequency and the -10 dB bandwidth.

1) Iteration 1: Basic L1 Patch

The initial design consisted of the microstrip patch without defects or slots, optimized for the L1-band. This baseline configuration achieved a simulated S_{11} parameter of -19.71 dB (1.524-1.56GHz) and a peak gain of 2.18 dB. While functional, the gain was limited by the inherent losses of the FR-4 substrate.

2) Iteration 2: Proposed Design (with DGS & Slots)

In the final proposed design, four symmetric corner slots and a Rectangular Defected Ground Structure (DGS) were incorporated. These modifications significantly improved the radiation characteristics. The simulation yielded an improved S_{11} parameter of -20.82 dB (1.54-1.577 GHz) and an enhanced peak gain of 2.38 dB.

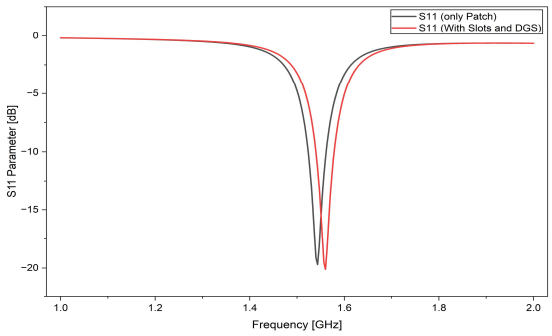


Fig. 2. Comparison of S_{11} Parameters

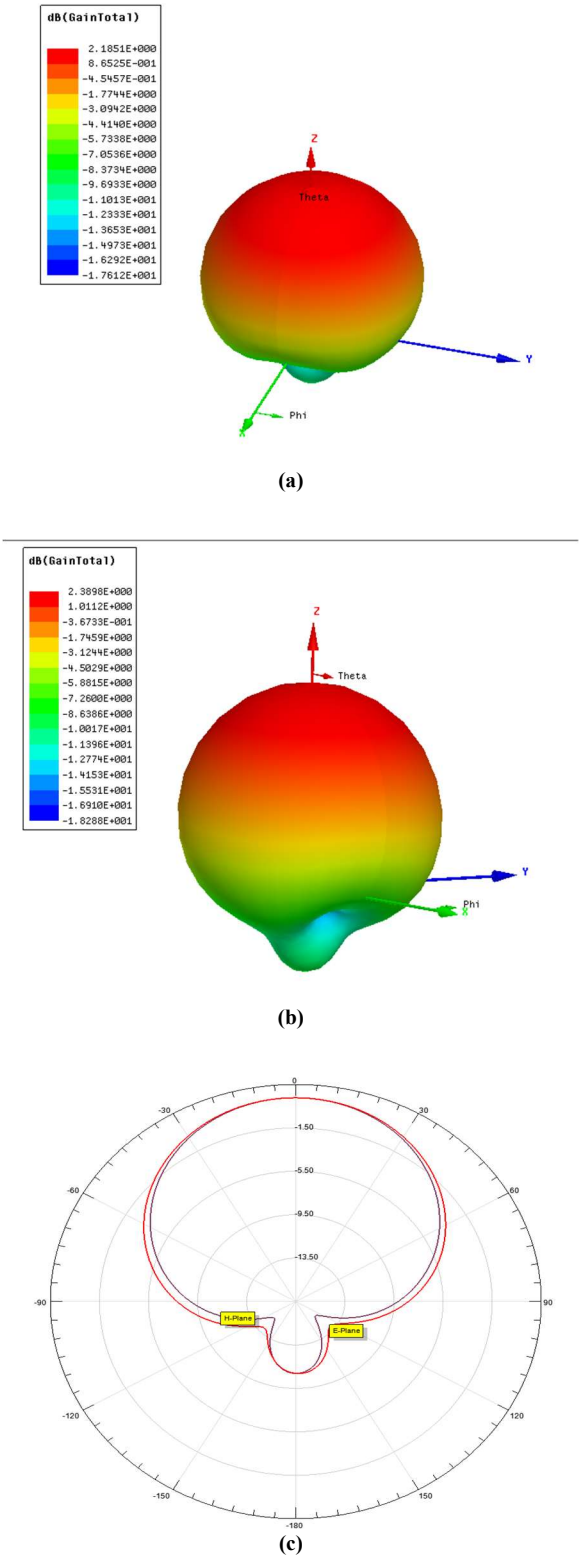


Fig. 3. Polar Gain Plots: (a) Only Patch; (b) Patch with Slots and DGS; (c) E-Plane and H-Plane Gain Plots

Table 2: Simulated Performance Comparison

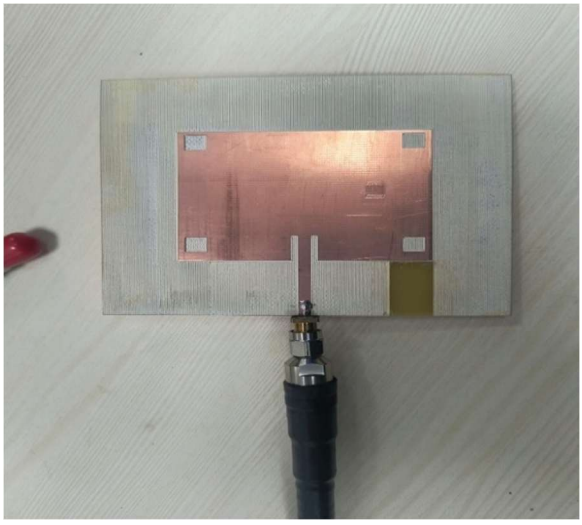
Iteration	Configuration	S_{11} Parameter [dB]	Peak Gain[dB]
Iteration 1	Basic Patch (L1)	-19.71	2.18
Iteration 2	Proposed (DGS + Slots)	-20.82	2.38

B. Mechanism of Gain Enhancement

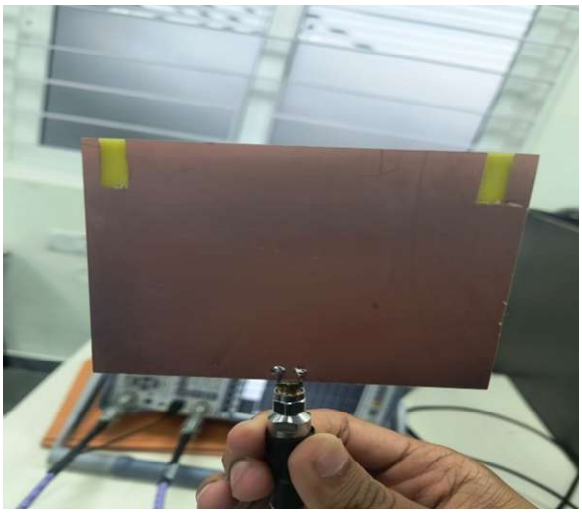
The observed improvement in gain is primarily attributed to the suppression of surface waves, which constitute a dominant loss mechanism in microstrip antennas fabricated on thick, high-permittivity FR-4 substrates as presented by Al-Falahy *et al.* [3], Alam *et al.*[4]. The implemented Rectangular Defected Ground Structure (DGS) functions as a spatial band-stop filter, effectively modifying the boundary conditions of the ground plane [5]. By perturbing the current distribution on the ground plane, the DGS inhibits the propagation of surface waves and minimizes edge diffraction scattering [6]. Consequently, electromagnetic energy that would otherwise be trapped within the substrate or scattered laterally is redirected into the main radiation lobe. This concentration of energy significantly enhances the antenna's directivity and radiation efficiency, resulting in the measured gain improvement despite the lossy nature of the substrate [7], [8]**.**

C. Experimental Validation

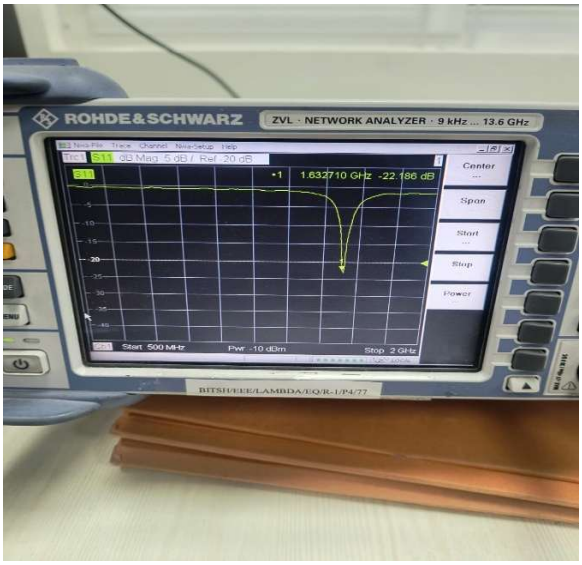
To validate the simulation results, the final proposed design was fabricated on a double-sided copper-clad FR-4 substrate. The reflection coefficient was measured using a Rohde & Schwarz ZVL Vector Network Analyzer (VNA). The measured results demonstrated an S_{11} of -22.186 dB at a resonant frequency of 1.632 GHz. As shown in Fig. 4(c).



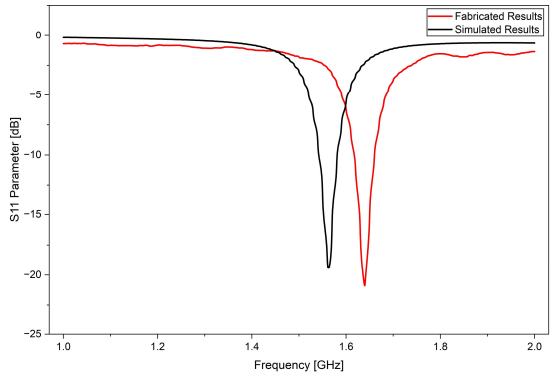
(a)



(b)



(c)



(d)

Figure 4: Fabricated Antenna: (a) Patch; (b) Ground Plane; (c) VNA measurements; (d) Comparison between Fabricated and Simulated Antenna.

A frequency upward shift of approximately 0.06 GHz (60 MHz) was observed between the simulated (1.575 GHz) and measured (1.632 GHz) resonance. This deviation is attributed to fabrication tolerances and variations in the dielectric constant (ε_r) of the commercial FR-4 substrate, which can typically vary between 4.3 and 4.8. Despite this shift, the antenna maintains a S_{11} parameter well below -10 dB, validating its functionality for embedded GNSS applications.

IV. COMPARISON WITH EXISTING WORKS

To evaluate the effectiveness of the proposed design, the simulated performance metrics were compared against existing GNSS antennas reported in recent literature. The comparison, summarized in Table III, highlights the trade-offs between gain, substrate complexity, and manufacturing cost. The designs by Savarkar *et al.* [4] and Ayad *et al.* [1] represent standard single-layer implementations on FR-4. While [10] had a gain close to 2dB at its L5 band, the notch-based design in [1] improved this to 2.27 dB for the L1 band. A more advanced multi-layer approach was presented by Li *et al.* [9], employing a stacked patch configuration with high-permittivity substrates (ε_{r1} = 6.15, ε_{r2} = 10.0). This complex structure achieved a high peak gain of 2.93 dBi at 1.575 GHz. In comparison, the Proposed Design achieves a competitive peak gain of 2.38 dB using a single layer of low-cost FR-4 (ε_r = 4.4). While slightly lower than the stacked ceramic design of [9], the proposed antenna offers a significantly simpler and more cost-effective fabrication process while outperforming other single-layer FR-4 solutions.

Table 3: Performance Comparison with Existing Works

Reference	Substrate Type	Frequency [GHz]	Technique	Peak Gain
Savarkar et al. [10]	FR-4 (Single)	1.176	Slot Cut (L5)	2dB
Ayad et al. [1]	FR-4 (Single)	1.575	Notched Slots	2.27dB
Li et al. [9]	Stacked (High-K)	1.575	Stacked Patches	2.93dBi
Proposed Work	FR-4 (Single)	1.575	DGS + Corner Slots	2.38dB

V. CONCLUSION

In this work, a gain-enhanced microstrip antenna was developed for the L1 GNSS frequency band. A primary focus of the investigation was overcoming the low radiation efficiency typically associated with thick FR-4 substrates. Through the strategic implementation of a Rectangular Defected Ground Structure (DGS), the design achieved a measured gain of 2.38dB.

The study confirmed that the DGS effectively suppresses surface wave propagation and reduces lateral scattering, thereby redirecting trapped energy into the broadside direction. Comparison between the simulated and measured S-parameters demonstrates a minimal frequency deviation, with a measured bandwidth of 37MHz (1.54-1.577 GHz). The proposed design successfully balances cost-effectiveness with high directivity, making it suitable for integration into consumer electronics and automotive navigation systems.

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